

1. Let  $A = \{1, 2, 3, 5\}$  and  $B = \{a, b, 2, c, 5\}$

a. Find  $A \cap B$

$$= \{2, 5\}$$

b. Find  $A \cup B$ .

$$= \{1, 2, 3, 5, a, b, c\}$$

c. Find  $B - A$ .

$$= \{a, b, c\}$$

d. Find  $A - B$ .

$$= \{1, 3\}$$

2. Let  $A$  be a nonempty subset of  $B$ . Which of the following is true?

~~(a)~~  $A \cap B = B$ .

~~(b)~~  $A \cup B = A$ .

(c)  $A \cup B = B$ .

~~(d)~~  $A \cap B = \emptyset$ .

3. Let  $A = \{3, 5\}$  and  $B = \{a, b, 5\}$ .

a. List all possible subsets of  $B$ .

$$\{\emptyset\}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, 5\}, \{b, 5\}, \{a, b, 5\}$$

b. Find  $A \times B$ .

$$\{(3, a), (3, b), (3, 5), (5, a), (5, b), (5, 5)\}$$

4. Prove that  $A \cup \emptyset = A$ .

*Proof.*  $\emptyset \subseteq A$  and  $A \subseteq A$  thus  $A \cup \emptyset \subseteq A$  and  $A \subseteq A \cup \emptyset$ .

Alternatively,

$(A \subseteq A \cup \emptyset)$  Let  $a \in A$  then  $a \in A$  or  $\emptyset$  thus  $a \in A \cup \emptyset$ .

$(A \cup \emptyset \subseteq A)$  If  $A = \emptyset$  then  $A \cup \emptyset = \emptyset \subseteq A$ . Otherwise if  $a \in A \cup \{\emptyset\}$  then  $a \in A$ . Thus  $A \cup \{\emptyset\} = A$ .  $\square$

5. Consider the map  $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$  where  $f(x, y) = xy$ . Prove that  $f$  is onto.

*Proof.* Let  $z \in \mathbb{R}$ .  $1 \in \mathbb{R}$  so then  $f(1, z) = 1 \cdot z = z$ . Thus  $f$  is onto.  $\square$

6. Define the relation  $\mathcal{R}$  on the real number set  $\mathbb{R}$  as  $x\mathcal{R}y \iff xy \geq 0$ .

a. Show that  $\mathcal{R}$  is reflexive and symmetric.

*Proof.* Reflexive:  $x \cdot x = x^2 \geq 0$ .

Symmetric: If  $x \cdot y \geq 0$  then  $x \cdot y = y \cdot x \geq 0$ .  $\square$

b. Is  $\mathcal{R}$  an equivalence relation? Explain.

No.  $\mathcal{R}$  is not transitive.  $-1 \cdot 0 \geq 0$  and  $0 \cdot 1 \geq 0$  but  $-1 \cdot 1 = -1 < 0$ .

7. Let  $X$  be a nonempty set. Define the following relation on the set of subsets of  $X$  as follows  $A\mathcal{R}B \iff A \cap B \neq \emptyset$  for  $A, B \subseteq X$ .

a. Show that  $\mathcal{R}$  is symmetric.

*Proof.* Suppose that  $A \cap B \neq \emptyset$  then  $B \cap A \neq \emptyset$ .  $\square$

b. Show that  $\mathcal{R}$  is not reflexive.

*Proof.*  $\emptyset \in X$ , and  $\emptyset \cap \emptyset = \emptyset$ . □

8. Let  $f : A \rightarrow B$  and  $g : B \rightarrow C$  be two functions.

a. If  $g \circ f$  is one-to-one then

~~(a)~~  $f$  is onto.

(b)  $f$  is one-to-one.

(c)  $g$  is one-to-one.

(d) Both  $f$  and  $g$  are one-to-one.

~~(e)~~ None of the above.

b. If  $|A| > |B|$  and  $A \neq \emptyset$  can  $f$  be one-to-one?

No. By the Pigeon Hole Theorem.

9. Let  $S_3$  be the symmetric group on three elements.

a. Which of the following is true?

(a)  $S_3$  has at least one non-abelian subgroup.

~~(b)~~ All proper subgroups of  $S_3$  are abelian.

~~(c)~~  $S_3$  is abelian.

~~(d)~~  $S_3$  is cyclic.

b. What is the order of  $S_3$  and  $S_4$ ?

$$|S_3| = 3! = 6$$

$$|S_4| = 4! = 24$$

10. Let  $f : G_1 \rightarrow G_2$  be a group homomorphism and let  $H \leq G_2$ . Recall that  $f^{-1}(H) = \{g \in G_1 | f(g) \in H\}$ .

a. Which of the following is true?

~~(a)~~  $f^{-1}(H)$  consists only of the identity of  $G_1$ .

(b)  $f^{-1}(H)$  is a subgroup of  $G_1$ .

~~(c)~~  $gf^{-1}(H) = f^{-1}(H)g$  for all  $g \in G_1$  (the left and right cosets are equal).

~~(d)~~  $f^{-1}(H)$  is not a subgroup of  $G_1$

b. Let  $a, b \in G_1$ . Does  $f(ab) = f(a)f(b)$ ?

Yes. Since  $f$  is homomorphic  $f(ab) = f(a)f(b)$  by definition.

11. Let  $G$  be a finite group of order 12 and  $g \in G$ .

a. The order of  $g$  cannot be:

(a) 8

~~(b)~~ 6

~~(c)~~ 4

~~(d)~~ 3

b. Let  $|H| = 6$ . Does Lagrange's theorem imply that  $H \leq G$ ?

No. Lagrange's Theorem is a necessary but not sufficient condition for a subset to be a subgroup. For example, the alternating group  $A_4$  has 12 elements but no subgroup of order 6.

12. Let  $G$  be a group of order 4. Which of the following is true?

~~(a)~~  $G$  is a cyclic group.

~~(b)~~  $G$  is  $\mathbb{Z}_2 \times \mathbb{Z}_2$ .

~~(c)~~  $G$  is  $\mathbb{Z}_4$ .

(d)  $G$  is abelian.

**13.** Let  $G = \{e, a_1, a_2, a_3, a_4, a_5\}$  be a group where  $e$  is the identity.

**a.** Find a subgroup of  $G$ .

The trivial group  $e$  is a subgroup of every group  $G$ .

**b.** Find a subgroup of  $G$  distinct from the group given for part **(a)**.

$\langle a_i \rangle \leq G$ , and  $\langle a_i \rangle \neq e$  for any  $1 \leq i \leq 5$ .

**c.** Can  $|a_2| = 5$ ?

No. By Lagrange's theorem  $|a_2| \mid |G|$  and 56

**14.** Let  $H$  and  $K$  be subgroups be non-trivial subgroups of  $G$ . Prove or disprove that  $H \cap K$  is a subgroup of  $G$ .

*Proof.*  $H \cap K \leq G$ . Let  $a, b \in H \cap K$ . Then  $b^{-1} \in H$  and  $K$  since  $b \in H$  and  $K$  and  $H, K \leq G$ . Thus  $ab^{-1} \in H \cap K$  because  $a \in H$  and  $K$  and subgroups are closed under their operation.  $\square$

**15.** Let  $R$  be a ring and  $a, b \in R$ .

**a.** Which statement is true?

~~(a)~~ If  $ab = 0$  then  $a = 0$  or  $b = 0$ .

~~(b)~~ If  $a^2 = 0$  then  $a = 0$ .

~~(c)~~ If  $a \neq 0$  and  $b \neq 0$  then  $ab \neq 0$ .

**(d)** None of the above.

**b.** Show that  $\mathbb{Q}$ , the set of rational numbers, is a ring under addition and multiplication.

*Proof.*  $\mathbb{Q}$  is an abelian group.  $\mathbb{Q} \subset \mathbb{R}$  so then associativity and distribution hold under multiplication. Thus  $\mathbb{Q}$  is a ring.  $\square$